Table 1: **Baseline Parameter Analysis** - Increasing parameters does not significantly improve baseline performance: We increased baseline backbones from 6 to 8 layers (∼30% more parameters), yielding only minor gains (<1%), while still trailing our 6-layer model. This further highlights that our performance gains stem from our novel architectural methodologies rather than simple model capacity.

(50-epoch)	Best-(MOTOR)			Avg.-(8-baselines)		
	Total score ↓	IHM ROC	IHM PRC	Total score ↓	IHM ROC	IHM PRC
6-layer	**4.945**	0.872	0.547	5.416	0.810	0.434
8-layer	4.953	**0.878**	**0.554**	**5.390**	**0.817**	**0.438**

Table 2: **Time-Conditioned Foresee (TCF) Objective Analysis - Ablation**: Our learning objective consists of (a) L_{next_time} and (b) $L_{foresee}$, where $L_{foresee}$ is trained to predict (c) multiple future tokens conditioned on their specific occurrence times. Result shows that TCF's effectiveness comes from time-conditioning future tokens, not merely predicting more tokens, and that both (predicting intervals) and (predicting future values) are indispensable.

Ablation conditions	Total score (↓)
Time Conditioned Foresee (a,b,c)	**4.762**
Remove L_{next_time} (b, c)	4.816
Use only single token (next token) for $L_{foresee}$ (a, c)	4.844
Multiple tokens but without time conditioning (a,b)	4.839
Simple Multiple token prediction (b)	4.886
NTP (None)	5.054

Table 3: **Pathology Focused Binning Analysis - Hyperparameter** λ: Weighted binning ($\lambda > 0$) consistently improves performance over quantile binning, with the best results around $\lambda = 1$. (Scores represent Total Score ↓).

Binning Analysis	$\lambda = 0$ (Quantile)	0.5	1 (Ours)	1.5	2
Total score (↓)	4.827	4.800	**4.762**	4.800	4.790
IHM AUROC	0.877	0.878	0.881	**0.885**	0.882
IHM AUPRC	0.580	0.584	**0.596**	0.587	0.583

Table 4: **Pathology Focused Binning Analysis - Bin count**: Under "Value Share," performance improves as the bin count increases up to 50. Conversely, without sharing, performance degrades beyond 10 bins due to increased data sparsity. (Scores represent Total Score ↓).

Number of bins	2	5	10 (Default)	20	50
No value share	4.930	4.790	**4.762**	4.797	4.830
Value share	5.208	5.095	4.985	4.906	**4.883**

Table 5: **Dual-Calendar RoPE Analysis - Calendrical components**: We analyzed the impact of calendrical periods (ranging from 5 minutes to 300 years). Performance significantly degrades when the calendar dimension is reduced below 20, whether by sparsifying while fixing endpoints ("Fix both ends") or by retaining only high-frequency periods ("Fix high-freqs"). These results underscore the importance of our proposed calendrical components. (Scores represent Total Score ↓).

PE analysis	Calendar dim					
	40 (Ours)	30	20	10	6	0 (RoPE)
Fix both ends	4.762	**4.755**	4.789	4.841	4.842	4.861
Fix high-freqs	4.762	**4.759**	4.767	4.802	4.855	4.861

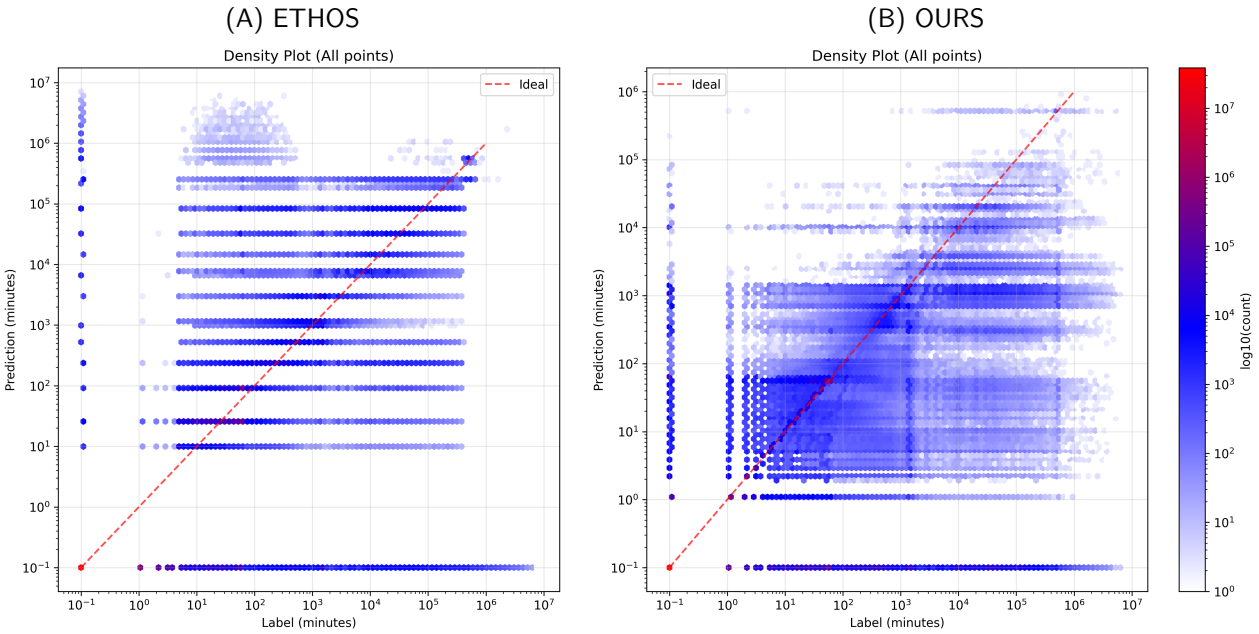

Figure 1: **Visualizing EHR generation time interval performance.** This figure illustrates the relationship between the actual next-token interval (x-axis) and the model-generated interval (y-axis) for (A) ETHOS and (B) our model. Our model significantly outperforms ETHOS within the clinically relevant time range (minutes to days), achieving performance improvements of up to 99%.